FACULDADE DE CIÊNCIAS E TECNOLOGIA
UNIVERSIDADE D COIMBRA

DEPARTAMENTO
DE ENGENHARIA
CIVIL

Erasmus Mundus WAVES / MEAECS

Written Report

COMPUTATIONAL MODELING WORK 2

Pranav Prakasan, UC 2024183416
Shahariar Ryehan, UC 2024161225

Work developed in the scope of the course
Computational Modeling
2024/2025

Scope/Context/Framework

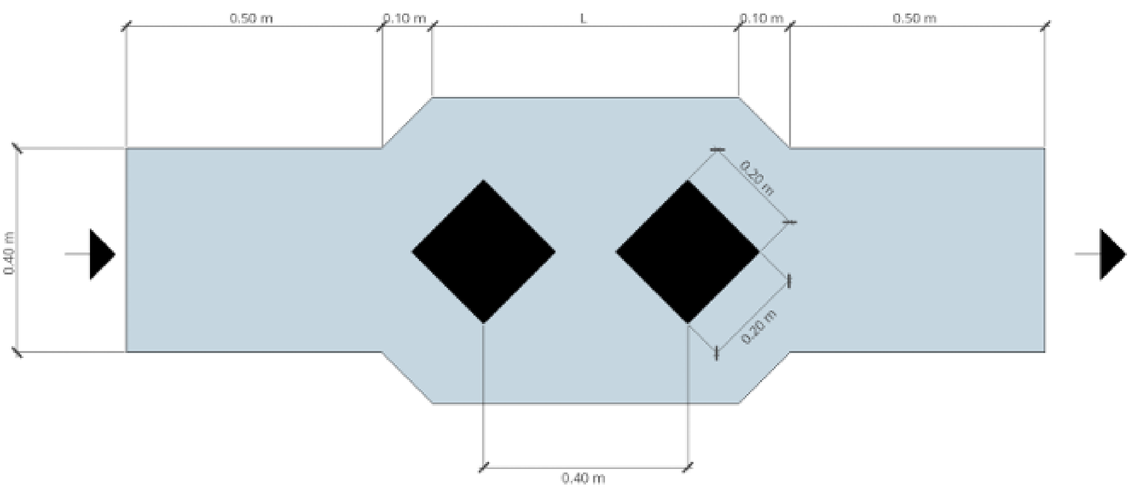
The aim of this report is to model, analyze, and evaluate the acoustic performance of a 2D attenuator system in the frequency domain using the Finite Element Method (FEM) with MATLAB's PDE Modeler. The study involves solving the Helmholtz equation to compute the sound pressure distribution and evaluate the Transmission Loss (TL) and Insertion Loss (IL) of the system. In the first part, the problem is parametrized by defining the PDE, boundary conditions, and surface absorption properties. A uniform mesh is created to ensure an accurate solution at the specified excitation frequency. The sound pressure distribution is then computed and visualized for its real, imaginary, and absolute values, along with the Sound Pressure Level (SPL). Next, the effect of mesh refinement is examined by starting with a coarser mesh and performing iterative refinements to observe the impact on pressure distribution and accuracy.

In the second part, the analysis is extended to octave frequency bands (125 Hz, 250 Hz, 500 Hz, 1000 Hz, and 2000 Hz). For each frequency band, the Transmission Loss and Insertion Loss are estimated by calculating the SPL at input and output boundaries for three frequencies within each band. The results are compared to assess the system’s acoustic performance across different frequencies, with consistent mesh usage to ensure reliable outcomes at higher frequencies. The report will provide insights into the acoustic behavior of the attenuator, its transmission and insertion losses, and the effects of mesh refinement on the accuracy of the simulation.

Given Parameters:

Based on the student's ID, with the last letter corresponding to A1=6 and B1=5, we have chosen Figure C for the analysis.

c) For A1=6 or A1=7



$$L = 0.6 + (A1 + B1)/100$$
$$L = 0.71$$

The Helmholtz equation governing the acoustic wave propagation in this system is given by:

$$\nabla^2 p + \left(\frac{\omega}{c}\right)^2 p = 0$$

- p is the acoustic pressure
- c is the sound propagation velocity
- ω is the angular frequency
- There is no air flow inside the device

The surface impedance is expressed as: $Z = \frac{p}{v} = \rho c \frac{1+\sqrt{1-\alpha}}{1-\sqrt{1-\alpha}} = \rho c \beta$

- α is the absorption coefficient
- ρ is the density of the medium

➢ The internal surfaces are rigid, governed by the Neumann condition ($v_n = 0$).

➢ The sound absorption coefficient for internal heterogeneities (valid from 100 Hz to 2500 Hz) is given by the formula:

$$\alpha = \frac{0.002f + 0.04\log(f) - (0.0008f)^2}{2\sqrt{A1 + 1}}$$

➢ A normal velocity of 1.0 mm/s is considered at the input.

➢ A perfect absorption condition is considered at the output.

Part A)

Considering an excitation frequency of: $f = 500 + A1 * 10 + B1\ (Hz)$

$$f = 565\ Hz$$

We calculated the frequency f, which allows us to compute the absorption coefficient α , surface impedance Z, the angular frequency ω and other related parameters needed for the analysis.

$$\alpha = 0.2228$$

$$\beta = 15.8870$$

$$\omega = 2\pi f \approx 3550\ rad/s$$

Z=6648.1

a1)

The domain was discretized using the "draw mode" to define the geometry based on the formula P1−P2−P3. Following this, the boundary conditions were applied in "boundary mode" for each section of the problem.

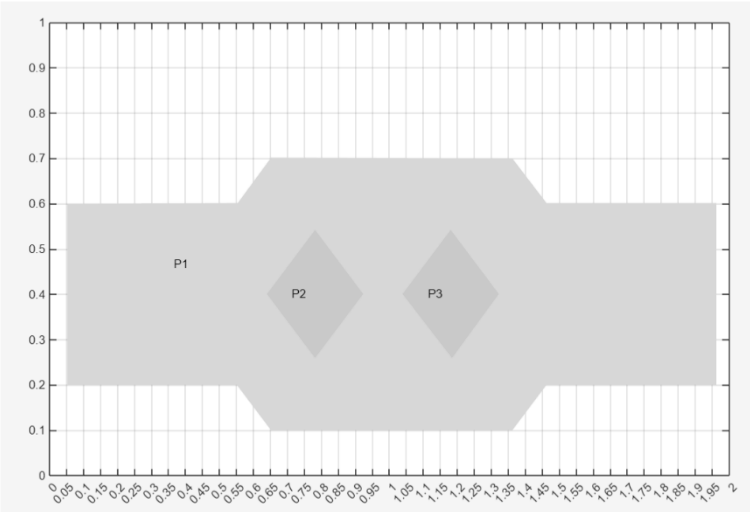
Left Side (Input Boundary)

At the input boundary, a normal velocity ϑ =0.001 m/s is imposed. The relationship between acoustic pressure and velocity is given by:

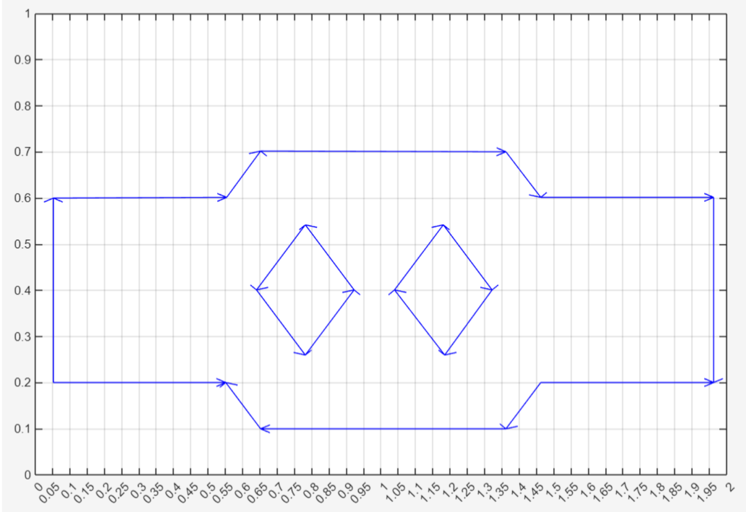
$$\vartheta = -\frac{1}{i\omega\rho}\frac{\partial p}{\partial x} \quad , \quad \frac{\partial p}{\partial x} = -\vartheta i\omega\rho$$

This forms the Neumann boundary condition for the input.

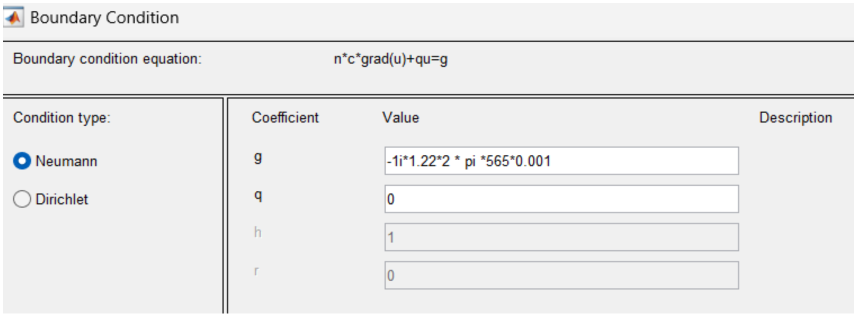
Draw mode



Boundary mode

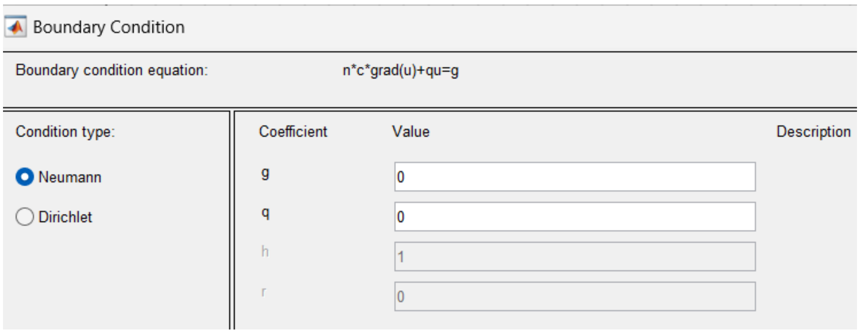


$$g = -1i * 1.22 * 2 * pi * 565 * 0.001$$



Top and Bottom Edges (outer boundaries):

These represent rigid walls, so apply a Neumann boundary condition with g=0 ,q=0 to simulate no normal velocity (perfect reflection).



Interior (Diamond Shapes):

Impedance Definition: $Z = \rho c \beta$

Relating Impedance with Pressure and Velocity: $Z = \frac{p}{v}$, $Zv - p = 0$

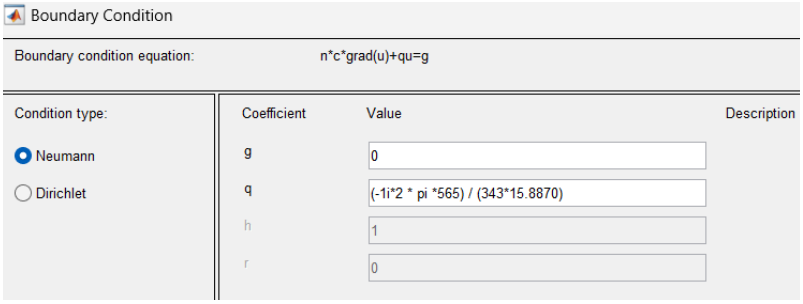
Boundary Condition from Impedance: From the relation $Zv - p = 0$, substitute $\vartheta = -\frac{1}{i\omega\rho}\frac{\partial p}{\partial x}$ (fluid velocity derived from pressure gradient):

$$Z\left(-\frac{1}{i\omega}\frac{\partial p}{\partial x}\right) - p = 0 \quad , \quad -\frac{\partial p}{\partial x} - \frac{p}{Z}i\omega\rho = 0$$

$$-\frac{\partial p}{\partial x} - \frac{p}{\rho c\beta}i\omega\rho = 0 \quad , \quad -\frac{\partial p}{\partial x} - \frac{p}{c\beta}i\omega = 0$$

This translates into the Neumann boundary condition as:

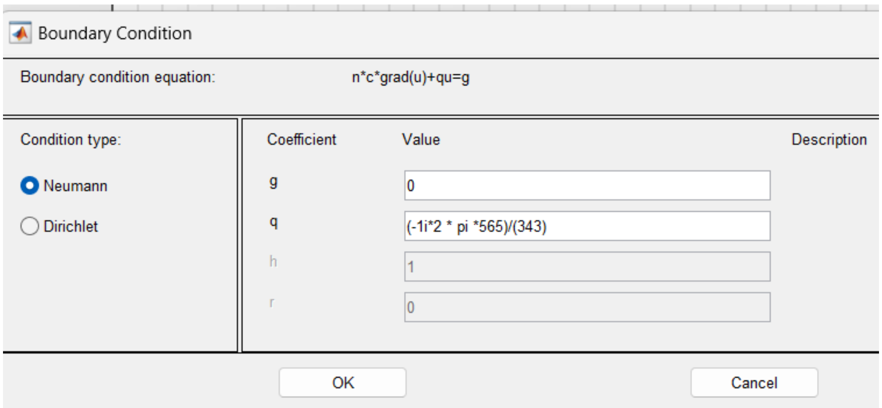
$$q = (-1i * 2 * pi * 565) / (343 * 15.8870)$$



Right side (output):

Where perfect absorption is required, the Robin boundary condition is appropriate. A perfectly absorbing boundary ensures no reflections, which can be expressed using the impedance boundary condition

$$q = (-1i * 2 * pi * 565)/(343)$$



Define the PDE

The problem involves solving the acoustic wave equation in the frequency domain. The equation can be expressed as:

$$-\nabla^2 p + k^2 p = 0$$

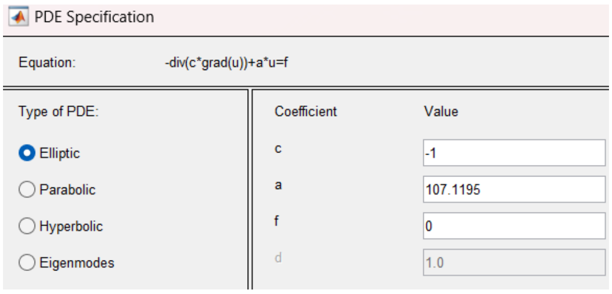
- p is the acoustic pressure.
- $k = \omega/c$,is the wave number , $k = \frac{565}{343} = 10.35$ rad/m.

where:

- ω=2πf , is the angular frequency
- c is the speed of sound in the medium.

In PDE Modeler:

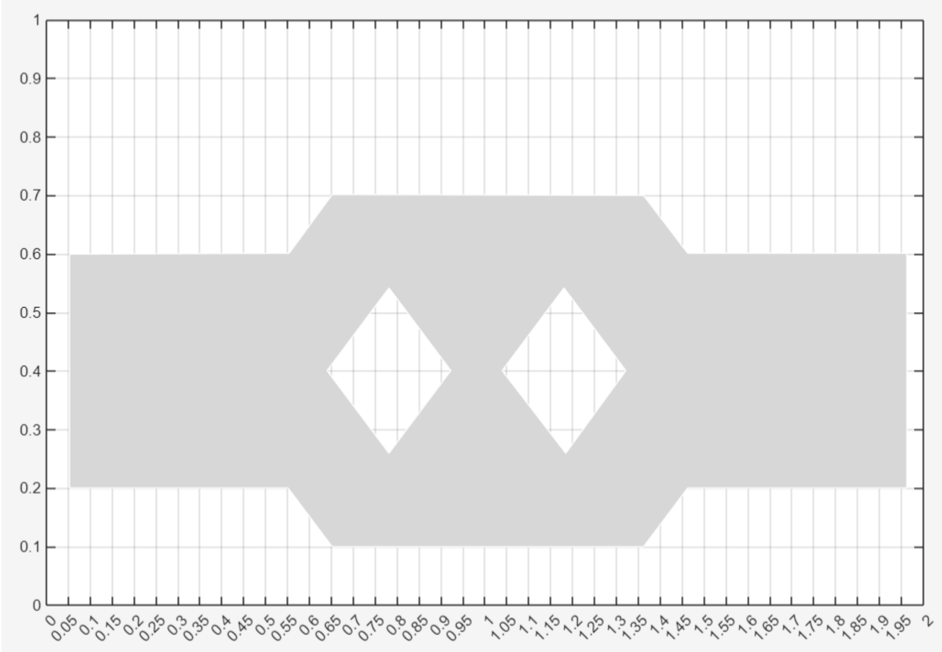
- This corresponds to an Elliptic PDE.
- PDE form: $-\nabla \cdot (c\nabla p) + ap = f$



where:

- c = −1 (for acoustic problems),
- $a = k^2 = (\omega / c)^2 = 107.1195$
- $f = 0$

PDE mode



a2)

The mesh resolution must be fine enough to accurately capture the acoustic wave propagation. A general rule for FEM in acoustic problems is:

Element size $(h) \leq \frac{\lambda}{6}$

where:

- λ is the wavelength of the acoustic wave, calculated as: $\lambda = \frac{c}{f} = \frac{343}{565} \approx 0.607\text{m}$

Thus, the maximum allowable element size is:

$$(h) \leq \frac{\lambda}{6} = \frac{0.607}{6} = 0.101m$$

To ensure accuracy and account for numerical dispersion, it is common to use $\frac{\lambda}{8}$ or $\frac{\lambda}{10}$ for higher accuracy. Here, we choose $\frac{\lambda}{10}$, which gives:

$$h = \frac{\lambda}{10} = \frac{0.607}{10} = 0.0607$$

- Mesh Size:Using uniform triangular or quadrilateral mesh with a maximum element size of h=0.0607m
- This resolution ensures at least 8 elements per wavelength, which provides a good balance between accuracy and computational cost.
- The exact number of elements depends on the total area of the geometry, but finer meshes will be concentrated near boundaries and regions with complex features

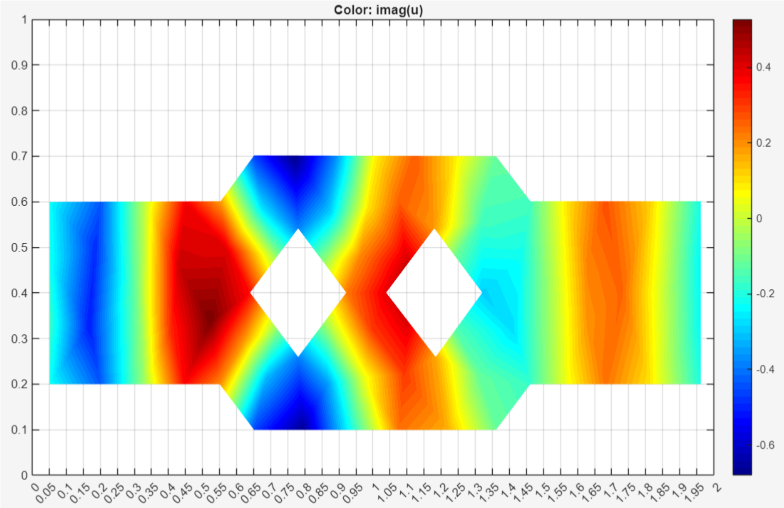
The adequacy of the mesh is further validated by the Helmholtz number (kh), where k is the wave number: $k = \frac{565}{343} = 10.35 \text{ rad/m}$.

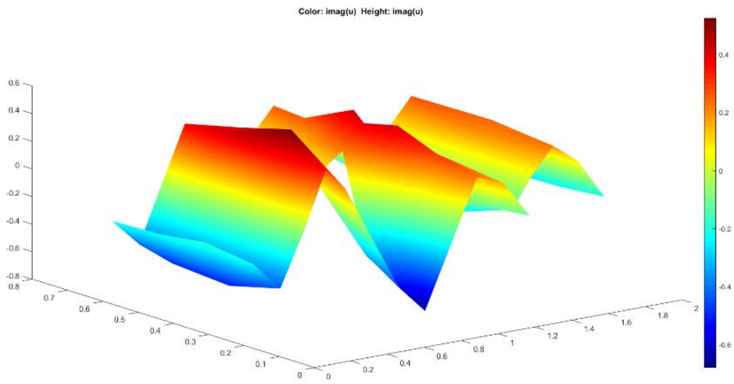
For a mesh size $h = 0.0607$ m , $kh = 10.35 * 0.076 \approx 0.63$ (dimensionless)

which is well below the standard threshold of $kh \leq 1$, ensuring low numerical dispersion.

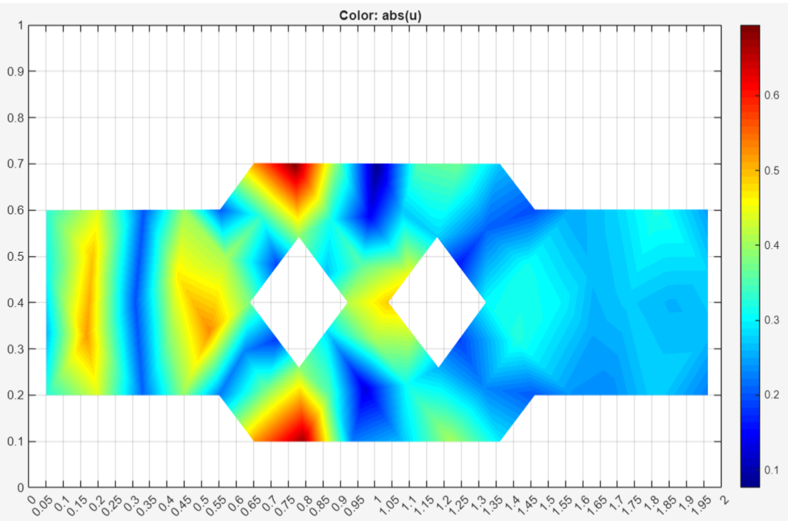
a3)

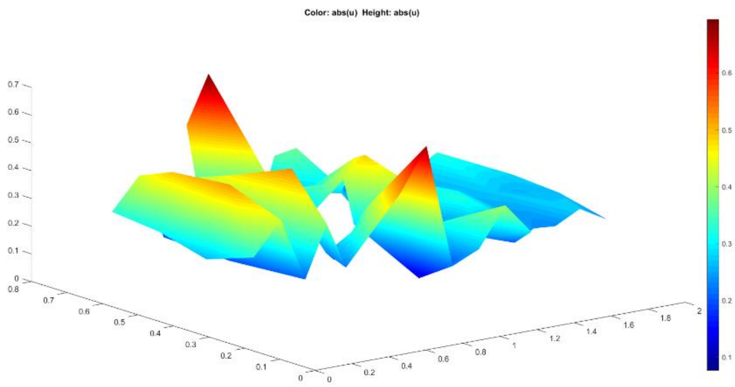
Imaginary values



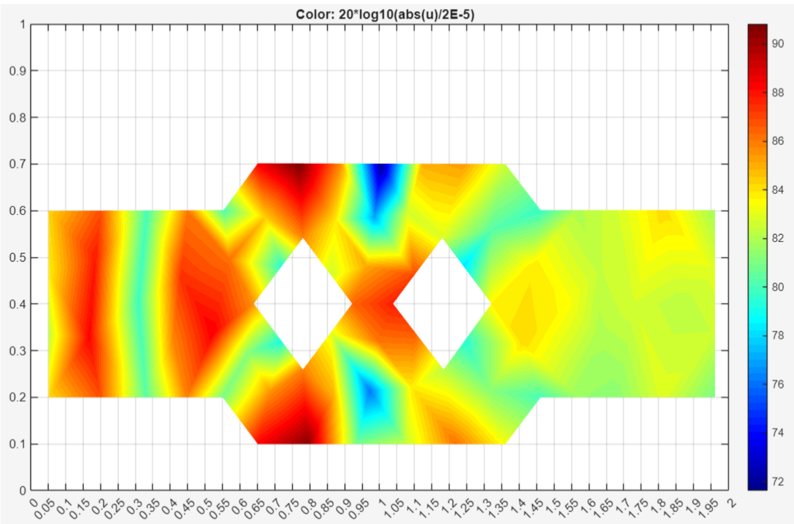


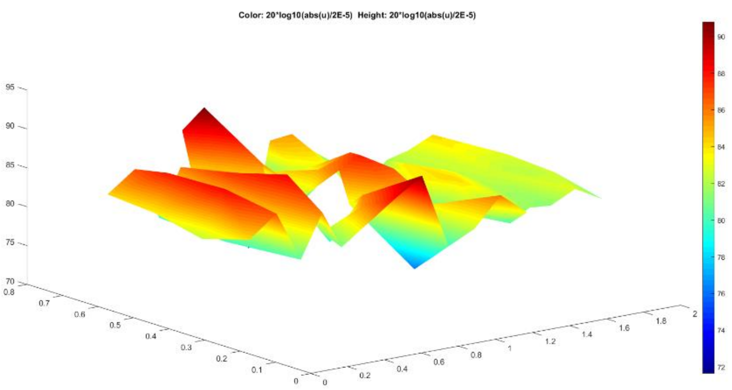
Absolute values



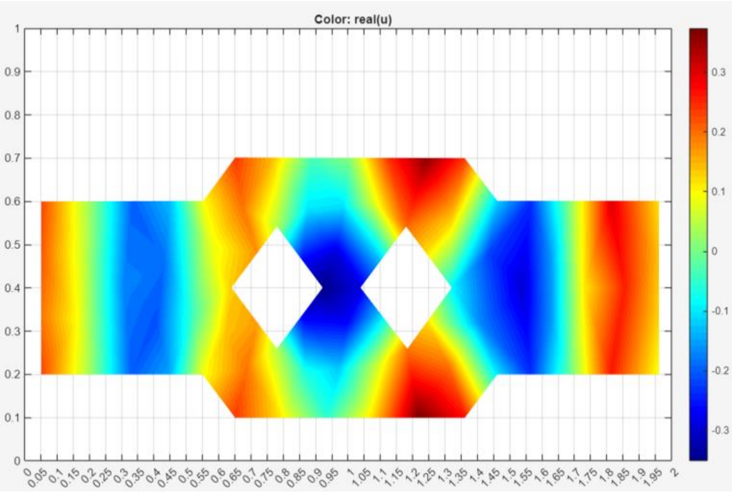


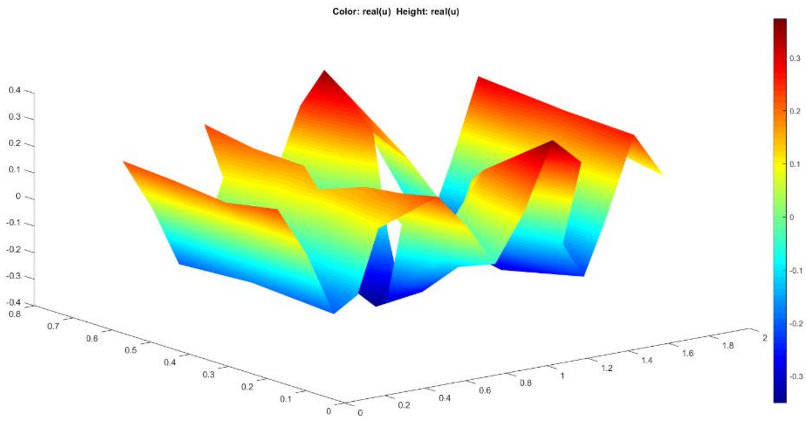
Sound Pressure Level (ref 2E-5Pa)





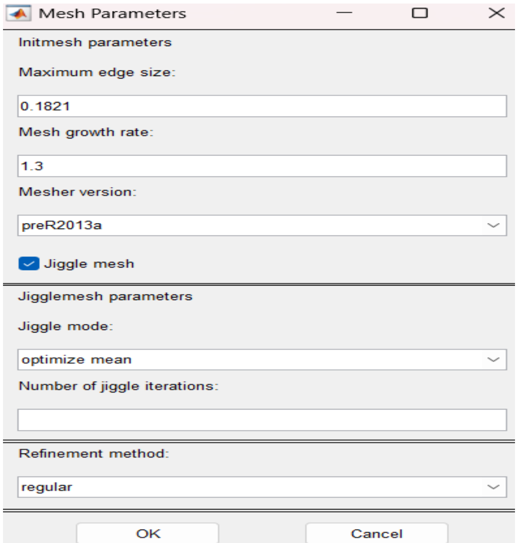
Real value





a4)

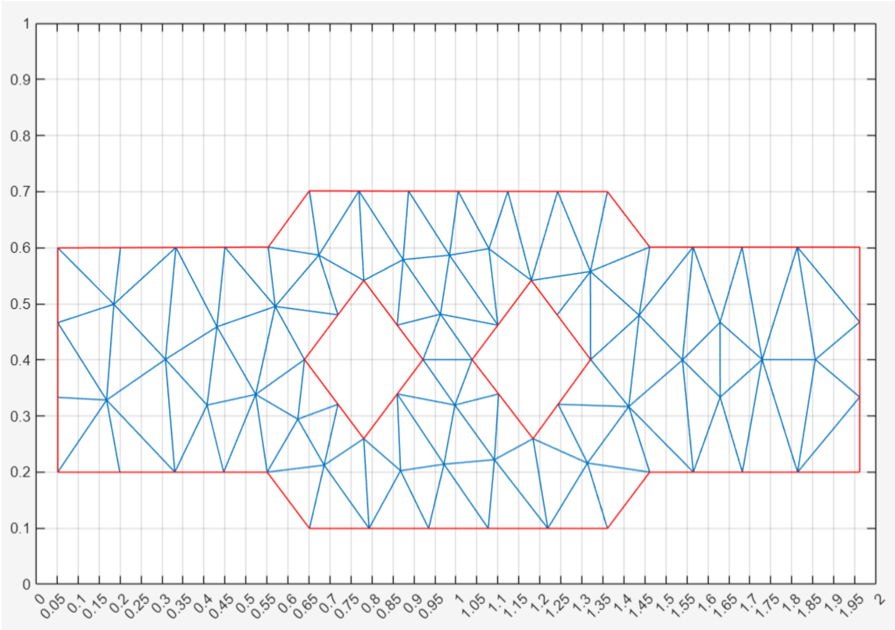
- Wavelength $\lambda = \frac{c}{f} = \frac{343}{565} \approx 0.607$
- For accuracy, element size $h = \frac{\lambda}{10} = \frac{0.607}{10} = 0.0607$
- Starting from a coarse mesh:
- Initial element size $= 3 \times h = 3 \times 0.0607 = 0.1821$

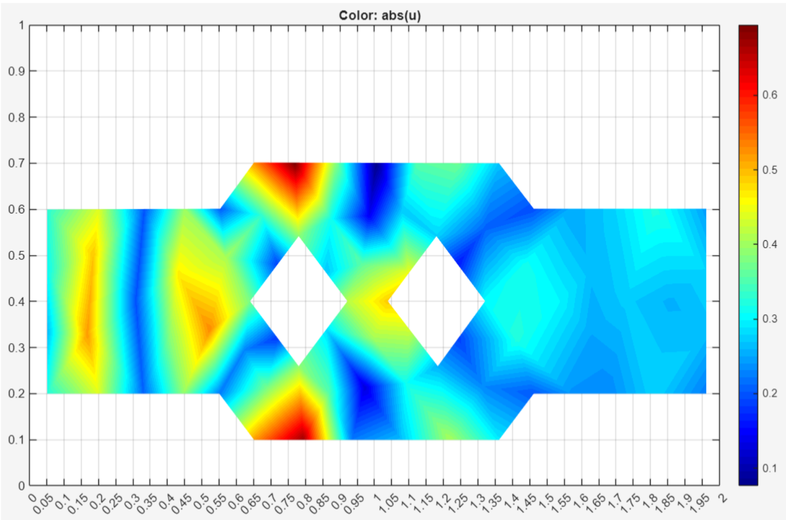


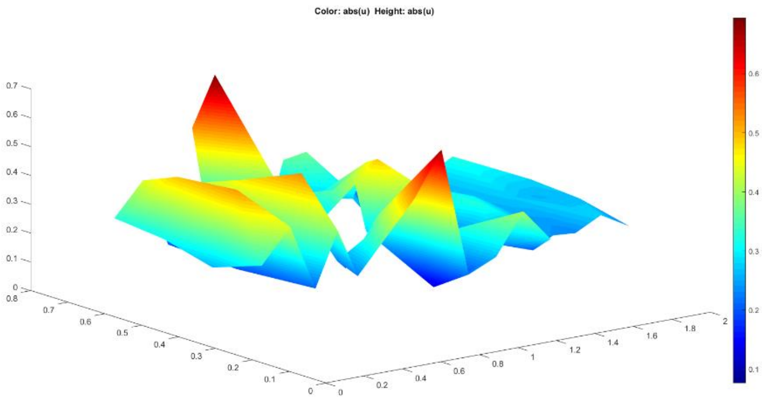
Starting with a coarse mesh having elements three times larger than the initial fine mesh (h=3×0.0607=0.1821) iterative refinement was conducted over five steps. The corresponding meshes had the following characteristics:

Mesh Step	Number of Nodes	Number of Triangles
Coarse Mesh	80	109
Step 1	270	436
Step 2	977	1744
Step 3	3699	6976
Step 4	14375	27904
Step 5	56655	111616

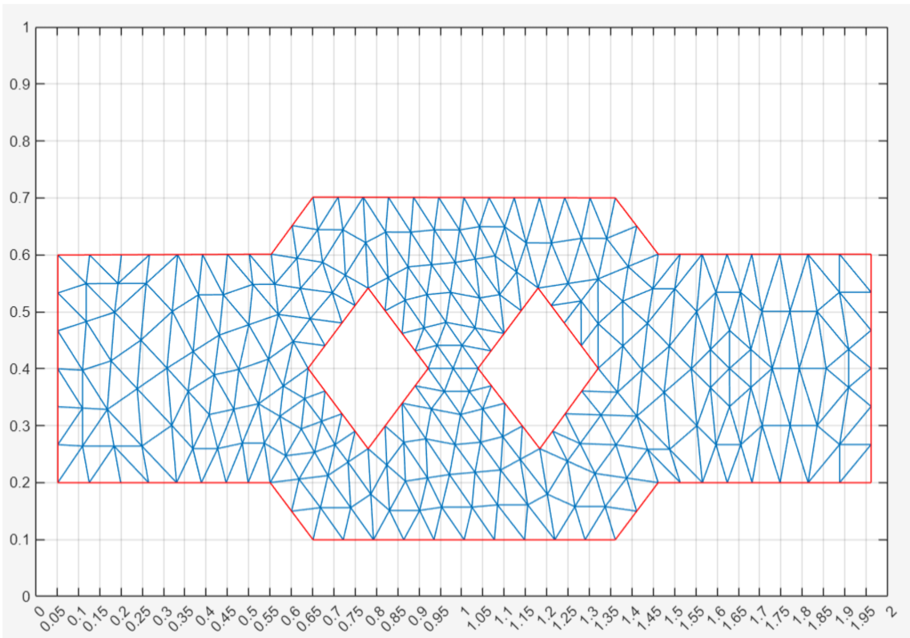
Coarse Mesh

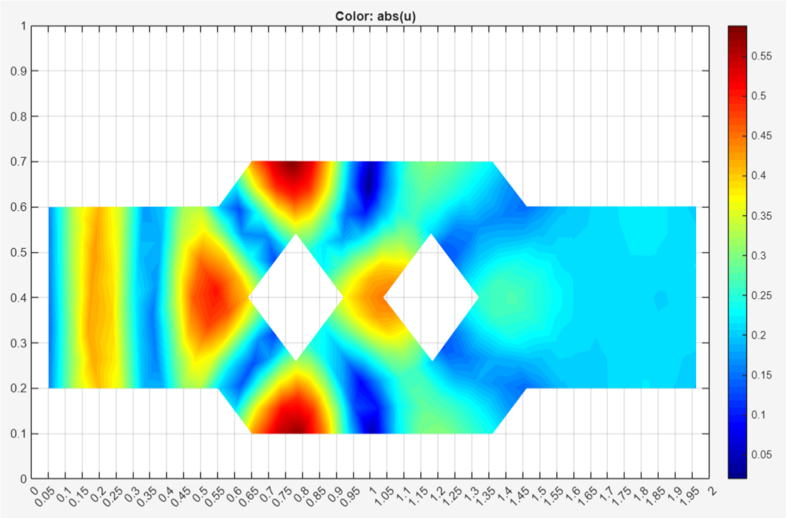


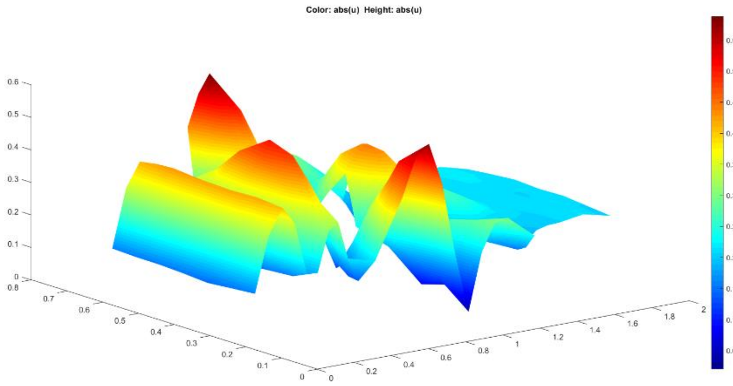




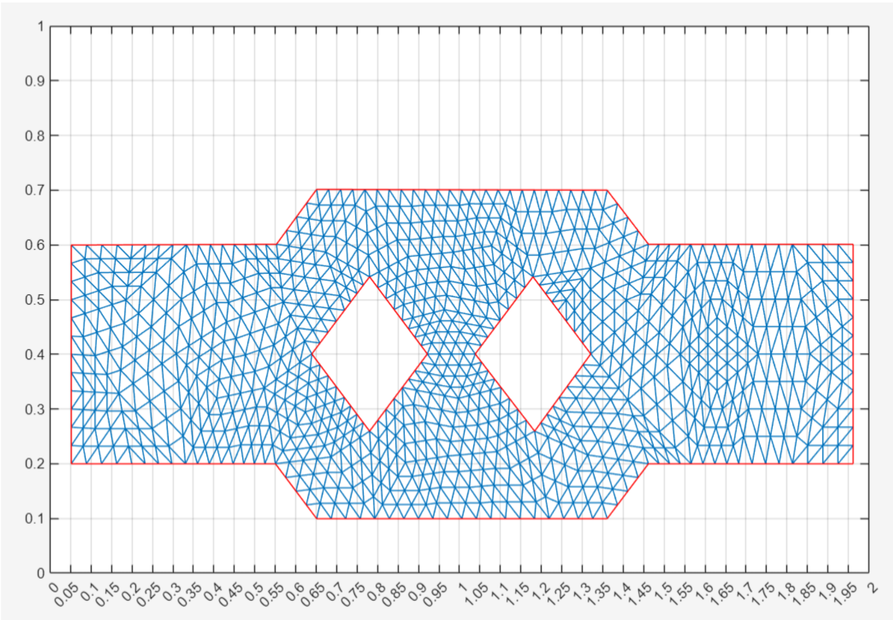
Step 1

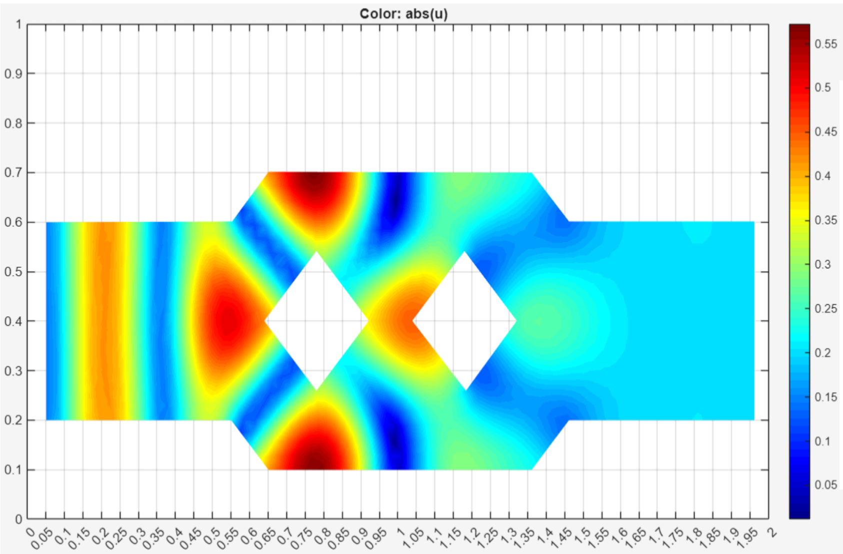


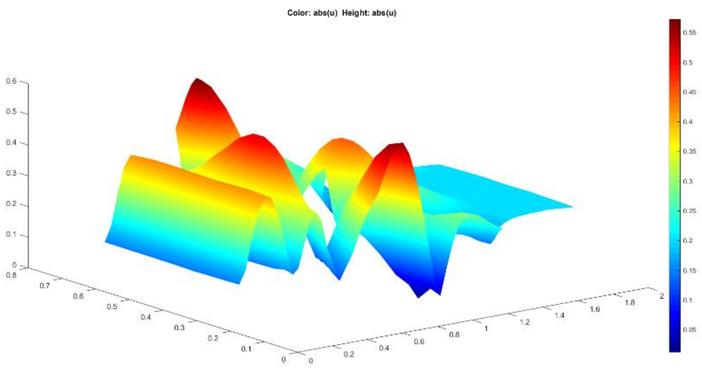

Color: abs(u)


Color: abs(u) Height: abs(u)

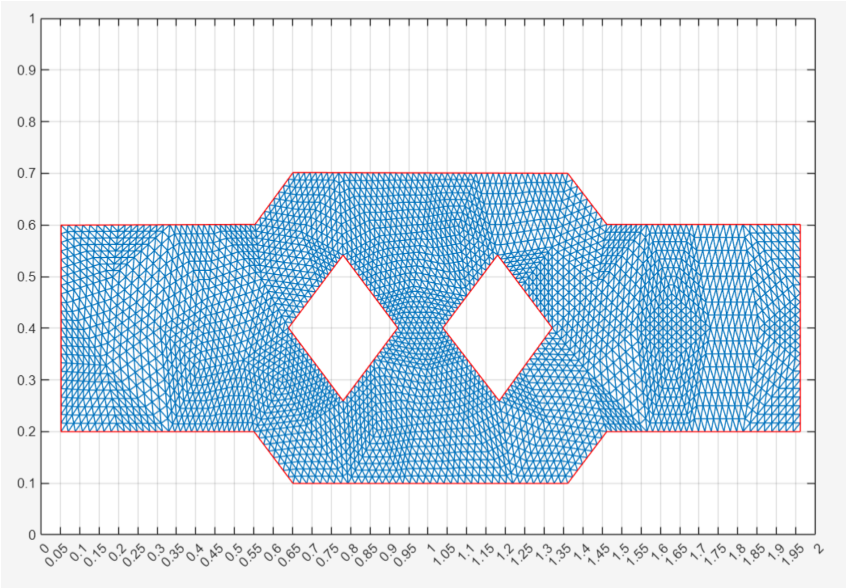
Step 2

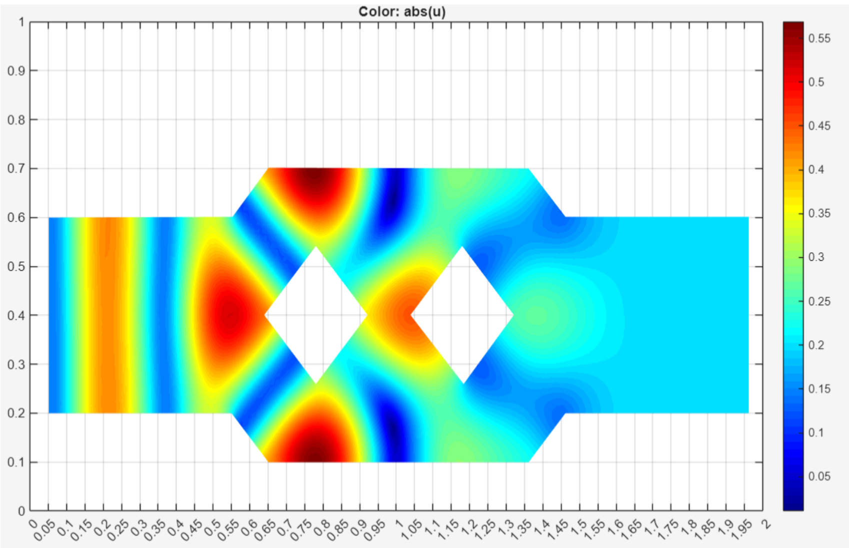


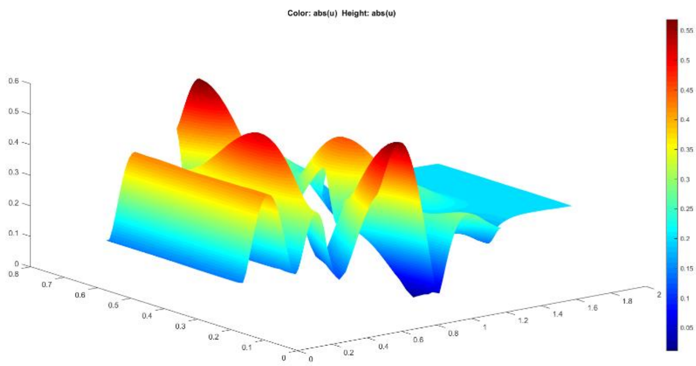

Color: abs(u)


Color: abs(u) Height: abs(u)

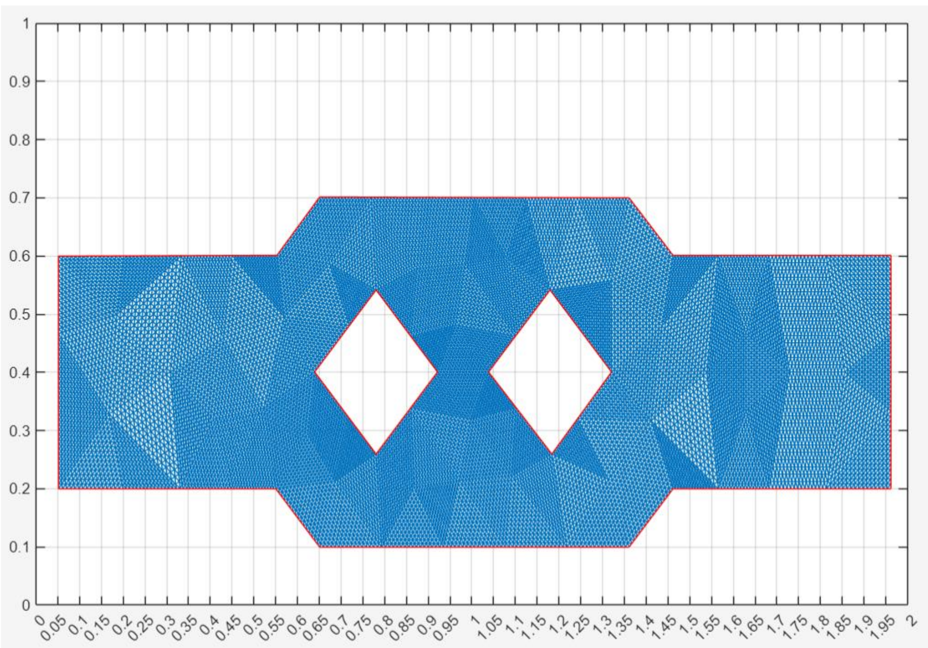
Step 3

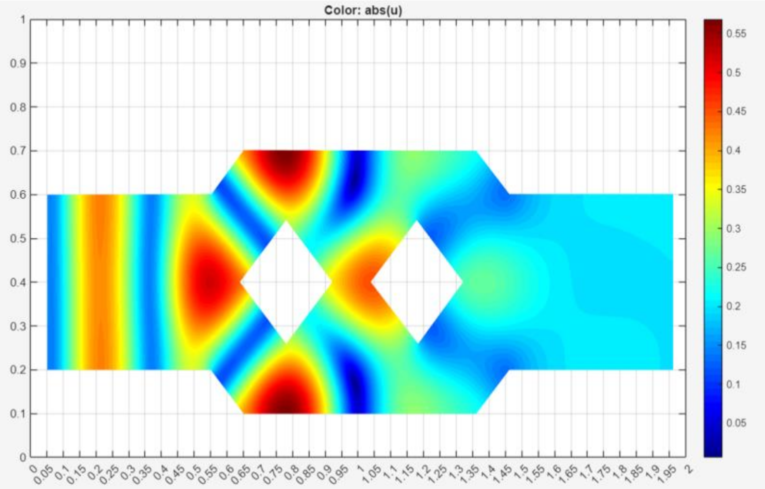


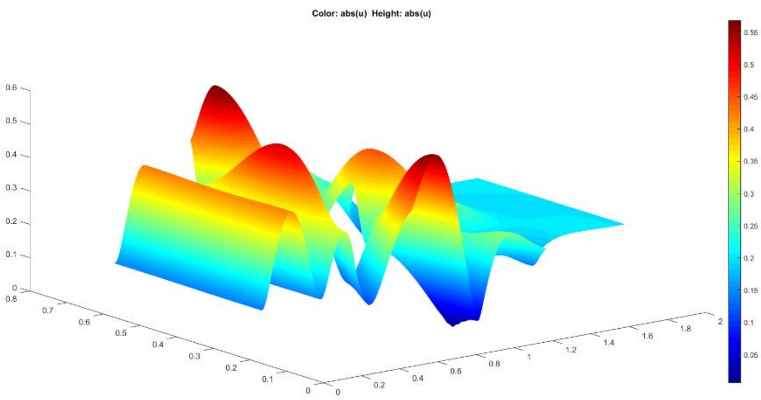




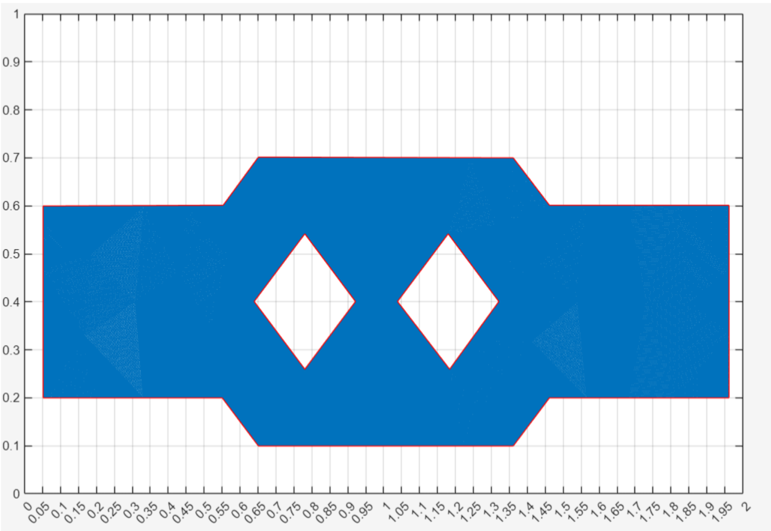
Step 4

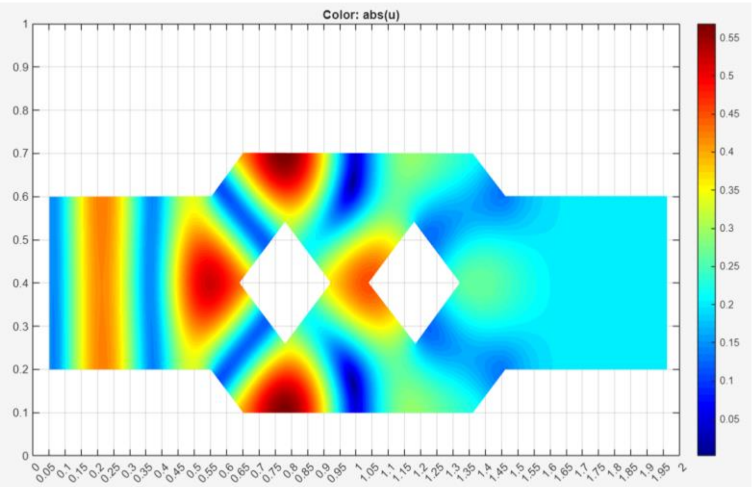


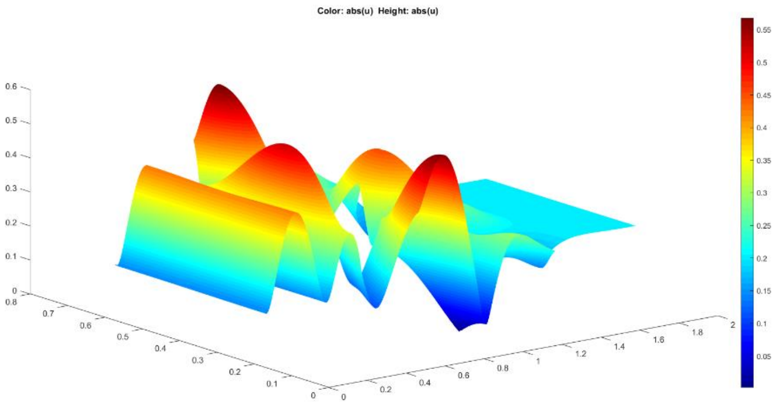




Step 5







The pressure distribution computed on the coarser mesh showed noticeable inaccuracies due to insufficient resolution, with jagged patterns and reduced fidelity in capturing the pressure gradients. As the mesh was refined, the results became progressively smoother, with the absolute values of pressure converging closer to the solution obtained with the finest mesh. The differences between subsequent refinements diminished significantly, indicating convergence.

Analyzing the figures, the element distribution improved with refinement, better capturing the wavelength $\left(\frac{\lambda}{10}\right)$ and reducing numerical dispersion errors. While the finer meshes (Steps 4 and 5) achieved higher accuracy with more nodes and elements, they also introduced a significant computational cost. Overall, the results highlight the trade-off between accuracy and efficiency, with finer meshes providing a more detailed and accurate pressure distribution at the expense of increased computational resources.

Part B)

The following table presents the Transmission Loss (TL) and Insertion Loss (IL) values for various octave frequency bands, calculated using sound pressure levels at different frequencies. The results are derived based on SPL measurements at the input and output boundaries, with and without the attenuator.

octave frequency bands	fi	fc	fs	Input With Attenuator				Output With Attenuator			
	$fc/2\text{^}(1/3)$		$fc * 2\text{^}(1/3)$	SPL fi	SPL fc	fs	Average	SPL fi	SPL fc	fs	Average
125	99.2	125	157.5	88.0	88.4	86.6	87.7	86.9	87.1	86.0	86.7
250	198.4	250	315.0	85.2	84.2	94.0	90.2	85.3	84.6	89.1	86.8
500	396.9	500	630.0	73.8	93.2	99.7	95.8	78.4	86.2	84.1	83.9
1000	793.7	1000	1259.9	82.0	88.3	90.2	88.0	68.5	73.1	82.4	78.3
2000	1587.4	2000	2519.8	87.0	92.6	90.5	90.6	88.2	82.3	87.6	86.7

Transmission Loss $TL = SPL(input, with\ attenuator) - SPL(output, with\ attenuator)$	Output Without Attenuator Average	Insertion Loss $IL = SPL(output, without\ attenuator) - SPL(output, with\ attenuator)$
1.0	86.4	-0.3
3.3	86.4	-0.4
11.9	86.4	2.5
9.7	86.4	8.1
3.9	86.4	-0.3

- The Transmission Loss (TL) for each frequency band was calculated by taking the difference between the Sound Pressure Level (SPL) at the input (with the attenuator) and the SPL at the output (with the attenuator). The TL values for each octave frequency band show how much sound energy is lost as the wave passes through the attenuating system. As we can observe from the table, the TL values vary across the frequency bands, with higher TL values observed at mid-range frequencies like 500 Hz (11.9 dB), indicating that the system attenuates more sound energy at these frequencies. For instance, at 1000 Hz, the TL value is 9.7 dB, while at lower frequencies such as 125 Hz, the TL is smaller (1.0 dB), suggesting that the system is less effective at attenuating low-frequency sound. The increase in TL from lower to higher frequencies reflects the system's ability to block or absorb sound more effectively at certain frequencies, likely due to the nature of the attenuating material and its interaction with sound waves.

- The Insertion Loss (IL) was computed by taking the difference between the SPL at the output without the attenuator and the SPL at the output with the attenuator. IL provides an indication of how much the attenuator contributes to reducing the sound energy at the output compared to a system without an attenuator. The IL values, shown in the table, generally show a reduction in sound energy when the attenuator is introduced. For example, at 500 Hz, the IL is 2.5 dB, meaning the attenuator provides a notable reduction in sound compared to the system without it. In contrast, negative IL values, such as at 125 Hz (-0.3 dB) and 2000 Hz (-0.3 dB), suggest that the attenuator had a negligible effect (almost equal to zero) or slightly increased the sound energy, possibly due to resonance effects.
- The SPL is averaged at specific points on the boundaries, and slight variations in the calculated pressure distribution could lead to differences in the averaged SPL values for systems with and without the attenuator. These variations can be attributed to the way sound pressure is distributed across the boundaries, particularly when using a numerical model with discretized meshes. Additionally, a low absorption coefficient (α) could contribute to the observed negative Insertion Loss (IL) values. In systems with low α, there is less attenuation of sound energy at the boundaries, leading to a smaller difference between the SPL at the output with and without the attenuator. Overall, the IL values show that the attenuator is most effective at frequencies where sound transmission is higher, and less effective at very low or very high frequencies, indicating the frequency-dependent performance of the system.